MSG No. 187.214 $P_c\bar{6}m2$ [Type IV, hexagonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-6m2`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 5: Wyckoff site: `2e`, site symmetry: `-6m2`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 6: Wyckoff site: `2f`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 7: Wyckoff site: `4g`, site symmetry: `3m.`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12]`
2	$[0, 0, -z]$	`[4,5,6,7,8,9]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[16,17,18,19,20,21]`

Table 8: Wyckoff site: `4h`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{1}{3},\, \frac{2}{3},\, -z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{1}{3},\, \frac{2}{3},\, z+\frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, \frac{1}{2}-z\right]$	`[16,17,18,19,20,21]`

Table 9: Wyckoff site: `4i`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{2}{3},\, \frac{1}{3},\, z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, z+\frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2}-z\right]$	`[16,17,18,19,20,21]`

Table 10: Wyckoff site: `6j`, site symmetry: `mm2`

No.	position	mapping
1	$[x,\, -x,\, 0]$	`[1,6,8,11]`
2	$[x,\, 2x,\, 0]$	`[2,4,9,12]`
3	$[-2x,\, -x,\, 0]$	`[3,5,7,10]`
4	$\left[x,\, -x,\, \frac{1}{2}\right]$	`[13,18,20,23]`
5	$\left[x,\, 2x,\, \frac{1}{2}\right]$	`[14,16,21,24]`
6	$\left[-2x,\, -x,\, \frac{1}{2}\right]$	`[15,17,19,22]`

Table 11: Wyckoff site: `6k`, site symmetry: `m'm2'`

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[1,11,18,20]`
2	$\left[x,\, 2x,\, \frac{1}{4}\right]$	`[2,12,16,21]`
3	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	`[3,10,17,19]`
4	$\left[x,\, 2x,\, \frac{3}{4}\right]$	`[4,9,14,24]`
5	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	`[5,7,15,22]`
6	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[6,8,13,23]`

Table 12: Wyckoff site: `12l`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,8]`
2	$[-y,\ x-y,\ 0]$	`[2,9]`
3	$[-x+y,\ -x,\ 0]$	`[3,7]`
4	$[x,\ x-y,\ 0]$	`[4,12]`
5	$[-x+y,\ y,\ 0]$	`[5,10]`
6	$[-y,\ -x,\ 0]$	`[6,11]`
7	$\left[x,\ y,\ \frac{1}{2}\right]$	`[13,20]`
8	$\left[-y,\ x-y,\ \frac{1}{2}\right]$	`[14,21]`
9	$\left[-x+y,\ -x,\ \frac{1}{2}\right]$	`[15,19]`
10	$\left[x,\ x-y,\ \frac{1}{2}\right]$	`[16,24]`
11	$\left[-x+y,\ y,\ \frac{1}{2}\right]$	`[17,22]`
12	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[18,23]`

Table 13: Wyckoff site: `12m`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,20]`
2	$\left[-y,\ x-y,\ \frac{1}{4}\right]$	`[2,21]`
3	$\left[-x+y,\ -x,\ \frac{1}{4}\right]$	`[3,19]`
4	$\left[x,\ x-y,\ \frac{3}{4}\right]$	`[4,24]`
5	$\left[-x+y,\ y,\ \frac{3}{4}\right]$	`[5,22]`
6	$\left[-y,\ -x,\ \frac{3}{4}\right]$	`[6,23]`
7	$\left[-x+y,\ -x,\ \frac{3}{4}\right]$	`[7,15]`
8	$\left[x,\ y,\ \frac{3}{4}\right]$	`[8,13]`
9	$\left[-y,\ x-y,\ \frac{3}{4}\right]$	`[9,14]`
10	$\left[-x+y,\ y,\ \frac{1}{4}\right]$	`[10,17]`
11	$\left[-y,\ -x,\ \frac{1}{4}\right]$	`[11,18]`
12	$\left[x,\ x-y,\ \frac{1}{4}\right]$	`[12,16]`

Table 14: Wyckoff site: `12n`, site symmetry: `.m.`

No.	position	mapping
1	$[x,\ -x,\ z]$	`[1,11]`
2	$[x,\ 2x,\ z]$	`[2,12]`
3	$[-2x,\ -x,\ z]$	`[3,10]`
4	$[x,\ 2x,\ -z]$	`[4,9]`
5	$[-2x,\ -x,\ -z]$	`[5,7]`
6	$[x,\ -x,\ -z]$	`[6,8]`
7	$\left[x,\ -x,\ z+\frac{1}{2}\right]$	`[13,23]`
8	$\left[x,\ 2x,\ z+\frac{1}{2}\right]$	`[14,24]`
9	$\left[-2x,\ -x,\ z+\frac{1}{2}\right]$	`[15,22]`
10	$\left[x,\ 2x,\ \frac{1}{2}-z\right]$	`[16,21]`
11	$\left[-2x,\ -x,\ \frac{1}{2}-z\right]$	`[17,19]`
12	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[18,20]`

Table 15: Wyckoff site: 24o, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x-y,\ z]$	[2]
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x,\ x-y,\ -z]$	[4]
5	$[-x+y,\ y,\ -z]$	[5]
6	$[-y,\ -x,\ -z]$	[6]
7	$[-x+y,\ -x,\ -z]$	[7]
8	$[x,\ y,\ -z]$	[8]
9	$[-y,\ x-y,\ -z]$	[9]
10	$[-x+y,\ y,\ z]$	[10]
11	$[-y,\ -x,\ z]$	[11]
12	$[x,\ x-y,\ z]$	[12]
13	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\ x-y,\ z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[18]
19	$\left[-x+y,\ -x,\ \frac{1}{2}-z\right]$	[19]
20	$\left[x,\ y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[-y,\ x-y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[23]
24	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	[24]